Department of Computer Science & Engineering

Software Engineering

UE24CS341A

Component Modeling & Architectural Pattern Selection

Name: Devansh Gupta	SRN: PES1UG24CS145	Section: C

Section 1

Section 1: Architecture Selection & Justification

Task: To evaluate the architectural styles and justify the choice for Problem Statement #19: Personalized Meal & Diet Subscription Manager.

Architecture Selection: I chose the Layered Architecture for the Personalized Meal & Diet Subscription Manager System.

Reason 1 (separation of concerns): presentation (Subscriber and Dietitian portal), business logic (menu generation, scheduling, delivery, payment) and data storage are cleanly separated, so each layer can be built, tested and audited on its own.

Reason 2 (fit to the workload): the system is a predictable request and response service (build a weekly menu, edit a subscription, check delivery) that maps naturally onto Presentation, Business and Data layers, giving a simple, low cost deployment for one clinical provider.

Security Advantage: the Data Layer is reachable only through the Business Layer, so sensitive metabolic data (allergens, diabetic limits, calories) is never accessed directly by the UI; one controlled Database Interface centralizes authentication, validation and auditing.

Performance Benefit: each layer is tuned independently (for example caching menu lookups in the Business Layer and moving the Data Layer to a stronger/better server), keeping weekly menu generation responsive.

Section 2

Section 2: Components, Interfaces & Component Diagram

Task: To identify components and interfaces and model them with provided interface (ball) and dependency notation.

Components: User Interface Component (Presentation); Meal Plan Manager, Payment Service and Delivery Tracking Components (Business); and Database Component (Data). The Meal Plan Manager also handles subscription scheduling (pause, skip and reschedule up to 12h before dispatch).

Key Interfaces: Meal Plan Interface, Payment Interface, Delivery Interface and Database Interface, all provided (ball) notation; layers are joined by dependencies from the User Interface to the Meal Plan Manager and on to the Payment, Delivery and Database components.

Fig-1: Component Diagram

